

Problema 4 (a) Determinați o formulă de cuadratură de tip Gauss de forma

$$\int_{-\infty}^{\infty} \frac{e^{-t}}{(1+e^{-t})^2} f(t) dt = A_1 f(t_1) + A_2 f(t_2) + R(f)$$

care să aibă grad maxim de exactitate. (3p)

(b) Aproximați integrala

$$\int_{-\infty}^{\infty} \frac{e^{-t}}{(1+e^{-t})^2} (x + \cos x + 1) dt,$$

cu 6 zecimale corecte folosind o cuadratură de tip Laguerre. (1p)

(corectie: $t + \cos(t) + 1$ la b)

clear

Solutie

Primul subpunct

Fie funcția pondere $w(t) = \frac{e^{-t}}{(1+e^{-t})^2}$. Observăm că $w(-t) = \frac{e^t}{(1+e^t)^2} = \frac{e^t}{e^{2t}(1+e^{-t})^2} = \frac{e^{-t}}{(1+e^{-t})^2} = w(t)$, deci

funcția este pară.

```
syms t;
w=exp(-t)/(1+exp(-t))^2
```

w =

$$\frac{e^{-t}}{(e^{-t} + 1)^2}$$

Dorim să găsim polinomul π_2 ortogonal pe $[-\infty, \infty]$ pentru a atinge gradul maxim de exactitate. Deoarece polinomul are 2 rădăcini (t_1, t_2) , polinomul va avea gradul 2.

Deoarece ponderea este pară, pentru ca $\int_{-\infty}^{\infty} w(t)\pi_2(t)t^{2k}dt$ să fie egal cu 0, deoarece $w(t)t^{2k}$ este o funcție pară (par * par = par), este necesar ca $\pi_2(t)$ să fie o funcție impară $\Rightarrow$ rădăcinile lui sunt fie egale cu 0, fie simetrice, în orice caz, $t_1 = -t_2 = a$, deci $\pi_2(t) = (t-a)(t+a) = t^2 - a^2$.

$$\pi_2 \perp 1 \Rightarrow \int_{-\infty}^{\infty} w(t)\pi(t)dt = 0 \Rightarrow \int_{-\infty}^{\infty} \frac{e^{-t}}{(1+e^{-t})^2} (t^2 - a^2)dt = 0$$

```
syms a;
int(w * (t^2 - a^2), t, -inf, inf)
```

ans =

$$\int_{-\infty}^{\infty} \left(-\frac{e^{-t} (a^2 - t^2)}{(e^{-t} + 1)^2} \right) dt$$

MATLAB-ul nu ne ajuta, vom rezolva in Wolfram Alpha:

$$\int_{-\infty}^{\infty} \frac{e^{-t} (t^2 - a^2)}{(1 + e^{-t})^2} dt = \frac{1}{3} (\pi^2 - 3a^2)$$

$$\frac{1}{3} (\pi^2 - 3a^2) = 0 \Leftrightarrow a^2 = \frac{\pi^2}{3} \Leftrightarrow a = \pm \frac{\pi \sqrt{3}}{3} \Rightarrow \pi_2 = t^2 - \frac{\pi^2}{3}$$

```
clear a;
pi2=t^2-sym(pi)^2/3
```

pi2 =

$$t^2 - \frac{\pi^2}{3}$$

Ca alternativa scurta la ce e mai sus se poate folosi *orto_poly_sym.m*

```
pi2_like=simplify(orto_poly_sym(w,t,sym(-inf),sym(inf),2)) % tot cu
WolframAlpha se calculeaza integralele
```

pi2_like =

$$t^2 - \int_{-\infty}^{\infty} \frac{t^2 e^t}{(e^t + 1)^2} dt - \frac{t \int_{-\infty}^{\infty} \frac{t^3 e^t}{(e^t + 1)^2} dt}{\int_{-\infty}^{\infty} \frac{t^2 e^t}{(e^t + 1)^2} dt}$$

Avand polinomul, ne mai raman de aflat coeficientii A_1 si A_2 . Deoarece sunt 2 coeficienti, avem nevoie de primele 2 momente ale functiei:

$$\mu_0 = \int_{-\infty}^{\infty} w(t) t^0 dt = A_1 + A_2$$

```
int(w, -inf, inf)
```

ans = 1

Deci $A_1 + A_2 = 1$.

$$\mu_1 = \int_{-\infty}^{\infty} w(t) t^1 dt = A_1 \cdot a - A_2 \cdot a. \text{ Deoarece } w(t) t^1 \text{ este o functie impara (par * impar)}$$

$$\Rightarrow \mu_1 = 0 \Leftrightarrow a(A_1 - A_2) = 0 \Leftrightarrow A_1 = A_2.$$

Din ultimele 2 relatii obtinem $A_1 = A_2 = \frac{1}{2}$.

Deci cuadratura Gauss obtinuta este $\int_{-\infty}^{\infty} w(t)f(t)dt = \frac{f\left(\frac{\pi\sqrt{3}}{3}\right) + f\left(-\frac{\pi\sqrt{3}}{3}\right)}{2} + R(f)$.

```
[rest, gr_maxim]=rest_gauss_fara_f(w,-inf,inf,pi2,2);
simplify(rest)
```

ans =

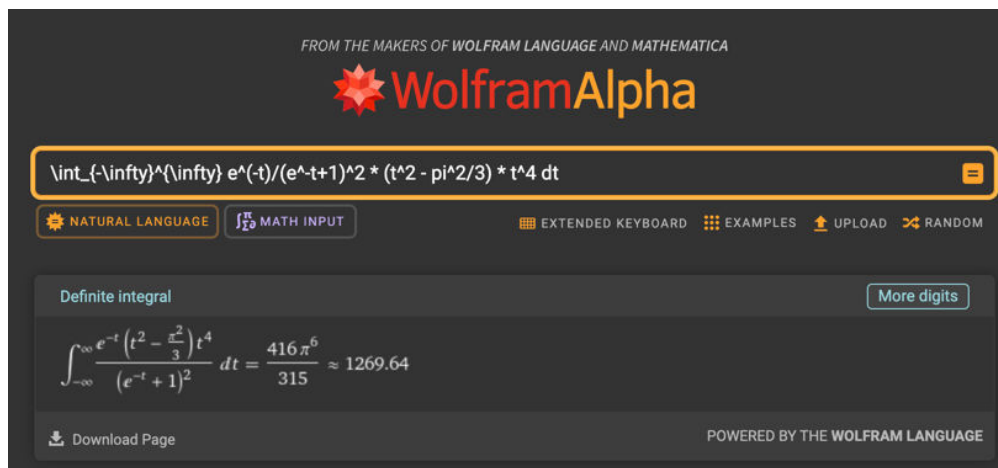
$$\frac{\int_{-\infty}^{\infty} \frac{e^{-t} (\pi^2 - 3t^2)^2}{(e^{-t} + 1)^2} dt}{216}$$

The screenshot shows the WolframAlpha interface. The input field contains the integral: $\int_{-\infty}^{\infty} e^{-t} (\pi^2/3 - t^2)^2 / (e^t + 1)^2 dt$. Below the input, there are tabs for "NATURAL LANGUAGE" and "MATH INPUT". The result is displayed as: $\int_{-\infty}^{\infty} \frac{e^{-t} (\frac{\pi^2}{3} - t^2)^2}{(e^t + 1)^2} dt = \frac{16\pi^4}{45} \approx 34.6343$. At the bottom, it says "POWERED BY THE WOLFRAM LANGUAGE".

Deci $R(f) = \frac{16\pi^4}{45 \cdot 24} f^{(4)}(\xi) = \frac{2\pi^4}{135} f^{(4)}(\xi)$, cu $\xi \in (-\infty, \infty)$.

Gradul maxim de exactitate este 3. Verificam:

The screenshot shows the WolframAlpha interface. The input field contains the integral: $\int_{-\infty}^{\infty} \frac{e^{-t} (t^2 - \pi^2/3) t^3}{(e^t + 1)^2} dt$. Below the input, there are tabs for "NATURAL LANGUAGE" and "MATH INPUT". The result is displayed as: $\int_{-\infty}^{\infty} \frac{e^{-t} (t^2 - \frac{\pi^2}{3}) t^3}{(e^t + 1)^2} dt = 0$. At the bottom, it says "POWERED BY THE WOLFRAM LANGUAGE".



Al doilea subpunct

$$I = \int_{-\infty}^{\infty} w(t)(t + \cos(t) + 1)dt$$

```
fn=matlabFunction(w*(t+cos(t)+1));
exp_quad=vpaintegral(fn, -inf, inf, 'RelTol', 1e-12, 'AbsTol', 0)
```

```
exp_quad = 1.27202905498
```

Cu schimbarea de variabila $y = -t$ obținem:

$$I = \int_{-\infty}^{\infty} w(-y)(-y + \cos(-y) + 1)dy = \int_{\infty}^{-\infty} w(y)(-y + \cos(y) + 1)dy$$

Adunand cu integrala initiala, obținem:

$$2I = \int_{-\infty}^{\infty} 2w(t)(\cos(t) + 1)dt \Leftrightarrow I = \int_{-\infty}^{\infty} w(t)(\cos(t) + 1)dt$$

Deoarece functia $\cos(t) + 1$ este para si $w(t)$ este para, produsul lor va fi o functie para, deci:

$$I = 2 \int_0^{\infty} w(t)(\cos(t) + 1)dt = 2 \int_0^{\infty} e^{-t} \cdot \frac{\cos(t) + 1}{(1 + e^{-t})^2} dt$$

Putem aplica o cuadratura Gauss-Laguerre.

```
n=18;
f=@(x) (cos(x)+1)./(1+exp(-x)).^2
```

```
f = function_handle with value:
@(x)(cos(x)+1)./(1+exp(-x)).^2
```

```
int=2*gauss_quad_num('Laguerre', f, n, 0)
```

```
int =
1.2720284186588
```

```
err=abs(exp_quad-int)
```

```
err = 0.00000063632332826936847636266004937366
```

Observam ca de la $n=18$ aproximatia noastra are 6 zecimale corecte.